REPORT 1

Data Cleaning Report

Data quality assessment, treatment and validation of the Rossmann datasets

Project	Rossmann Store Sales
Stage	Data Cleaning
Datasets	Train, Test, Store, Sample Submission
Report date	12 November 2026

1,017,209	0	0	13
Training rows	Missing cells after cleaning	Duplicate rows	Store columns after cleaning

1. Dataset Overview

Dataset	Rows	Columns	Duplicate Rows	Missing Cells
Train	1,017,209	9	0	0
Test	41,088	8	0	11
Store	1,115	10	0	2,343
Sample Submission	41,088	2	0	0

2. Missing Values Analysis

Dataset	Column	Missing Count	Missing (%)
Test	Open	11	0.03
Store	CompetitionDistance	3	0.27
Store	CompetitionOpenSinceMonth	354	31.75
Store	CompetitionOpenSinceYear	354	31.75
Store	Promo2SinceWeek	544	48.79
Store	Promo2SinceYear	544	48.79
Store	PromoInterval	544	48.79

3. Duplicate Analysis

Dataset	Duplicate Rows Found
Train	0
Test	0
Store	0

4. Missing Value Handling

- **CompetitionDistance:** median imputation, with a missingness indicator created.
- **Promo2 fields:** missing values handled as structurally not applicable.
- **Competition opening information:** handled using indicators.
- **Test Open:** missing values inferred using Store and DayOfWeek history.

5. Outlier Analysis

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	Outlier Count	Outlier (%)
Sales	4,859.00	8,360.00	3,501.00	-392.50	13,611.50	30,769	3.64
Customers	519.00	893.00	374.00	-42.00	1,454.00	40,853	4.84

- Outliers were detected using the IQR method.
- Genuine business outliers were retained.
- No automatic deletion or capping was performed.

6. Before and After Comparison

Dataset	Rows Before	Rows After	Cols Before	Cols After	Missing Before	Missing After	Dupes Before	Dupes After
Train	1,017,209	1,017,209	9	9	0	0	0	0
Test	41,088	41,088	8	8	11	0	0	0
Store	1,115	1,115	10	13	2,343	0	0	0

7. Final Data Quality Validation

Validation Check	Count	Status
Train missing cells	0	**PASS**
Test missing cells	0	**PASS**
Store missing cells	0	**PASS**
Train exact duplicates	0	**PASS**
Test exact duplicates	0	**PASS**
Store exact duplicates	0	**PASS**
Invalid train dates	0	**PASS**
Invalid test dates	0	**PASS**
Test Open missing	0	**PASS**
Negative Sales	0	**PASS**
Negative Customers	0	**PASS**
Invalid train DayOfWeek	0	**PASS**
Invalid test DayOfWeek	0	**PASS**

8. Cleaning Quality Summary

Dataset	Rows	Columns	Missing Values	Duplicate Rows
Train	1,017,209	9	0	0
Test	41,088	8	0	0
Store	1,115	13	0	0

9. Exported Cleaned Datasets

Datasets	`cleaned_train.csv`
	`cleaned_test.csv`
	`cleaned_store.csv`

10. Conclusion

The Rossmann datasets were cleaned and validated. Missing values, duplicates, data types, business rules and outliers were analysed. The cleaned datasets are ready for further Exploratory Data Analysis and Sales Prediction.